TBA Offseason Companion Wiki

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FMS Offseason Companion (BETA)

The TBA/FMS Offseaon Companion is a Docker container that runs on the FTA Toughbook, integrates with FMS, and performs data sync to TBA.

Note: *For 2025 Offseason Events, the Companion is in a limited beta. If you are interested in using this, reach out in FTA Slack for more.*

FTA Setup

Are you the FTA at an event using the Companion?

Follow the setup instructions here

User Guide

Are you a Scorekeeper using the Companion at an event?)

Refer the user guide here

Before The Event

You will need to bring with you:

- TBA Trusted API Keys for your event, which you can request here
- If your event is using FMS Cloud Sync, your FTA will have received the notepad token along with the event PIN. If you don't have a notepad token, reach out to FRC Support
- If your event is using Nexus AutoQueue, you'll need a Nexus API key. These can be generated automatically from the companion (if internet is available), otherwise follow the instructions before the event and bring the keys with you
- If your event will not have Internet access, follow the "Offline Setup" steps below, and bring the files with you on a USB drive

Before The Event: Offline Setup

if you will not have internet access at your event, you'll need to download the container images beforehand and bring them with you

1. Download and extract the latest release source code from the releases page (under the "Assets" drop down)
2. Download the `tba-offseason-companion-image-linux-amd64.tar` release artifact, as well. (If you have an Apple Silicon Mac, download `tba-offseason-companion-image-linux-arm64.tar` instead)
3. Optionally, download the `tba-offseason-companion-wiki.zip` release artifact, if you would like to bring a local copy of the documentation
4. Copy these files onto a flash drive and bring it with you to the event

At The Event: Container Setup

Depending on your internet availability, choose the corresponding instructions below. If you are running the Companion on a non-Windows machine, or prefer to set up the containers manually, there are more detailed steps at the bottom of this page.

With Internet (Automatic)

1. Download and extract the latest release source code from the releases page (under the "Assets" drop down)
2. Navigate to the extracted directory in Windows Explorer and run `start-docker.bat`

Without Internet (Automatic)

1. Copy the files you downloaded before the event from your USB drive onto the FTA Toughbook
2. Ensure you have extracted the latest release source code from this repository
3. Navigate to the extracted directory in Windows Explorer and run `offline-install.bat`. You will need to select the location of the downloaded container archive.

Companion Application Setup

The web UI will be available on the Toughbook at http://localhost:6001

Protip: You can get the IP Address of the Toughbook by running `ipconfig` in PowerShell. Then, you can navigate to `http://${TOUGHBOOK_IP}:6001` from the field server, if you'd like the Scorekeeper to operate the Companion

Authentication

On the main page of the companion, you will need to enter:

- your TBA Trusted API Key & Secret
- the event code used in FMS and the notepad token. If you created the event DB on FMS, then you can use `FTANOAPISYNC` as the token.

note: If you are using "legacy" FMS Auth instead of the Notepad token, then put the username in the Event Code field, and your API key in the "Notepad Token" field.

Manual Container Setup

If the automatic setup steps above don't work, or if you are planning to run the Companion on a non-windows platform, here are the steps

With Internet (Manual)

1. Download and extract the latest release source code from the releases page (under the "Assets" drop down)
2. Open the Docker Desktop application to initialize Docker (or ensure Docker is running on your system, however that works)
3. Open a Powershell terminal and navigate to that directory
4. Run `docker-compose up` to launch the Companion and navigate to http://localhost:6001

Without Internet (Manual)

1. Copy the files you downloaded before the event from your USB drive onto the FTA Toughbook
2. Open the Docker Desktop application to initialize Docker (or ensure Docker is running on your system, however that works)
3. Open a Powershell terminal and navigate to the location where you copied the files
4. Run `docker load --input .\tba-offseason-companion-image.tar`
5. Run `docker-compose up` to launch the Companion and navigate to http://localhost:6001

If you are running a playoff tournament using FMS test matches, how you assign match codes for TBA will depend on the playoff structure you are using. Ensure that the playoff type matches the format you expect to use, otherwise TBA will render the bracket incorrectly.

TBA Match Code Structure

A match code contains four parts:

- Tournament Level (`ef`, `qf`, `sf`, `f`)
- "Set" Number
- The letter `m`
- "Match Number"

Together, these parts form the match code, for example, `sf1m2`

Double Elimination Bracket

All the double elim matches are treated as `sfNm1`, where N is the FMS match number. The match number is always 1, unless there is a tie, in which subsequent replays increment the match number. In the official 8 alliance tournament, this results in 13 matches. In the 4 alliance bracket, this results in 5 matches.

The finals matches are treated as `f1mN`, where N is the finals match number (1 to 6, including overtime matches).

Round Robin

The round robin matches are treated as `sf1mN`, where N is the FMS match number (in the 6-alliance tournament, this is 15 matches).

The finals matches are treated as `f1mN`, where N is the finals match number (1 to 6, including overtime matches).

Single Elimination Bracket

Each "matchup" is its own "set", and the best-of-3 matches increment the match number from 1-3 (or more of there are ties).

Depending on the number of alliances, start with the corresponding comp level:

- 16 alliances: `ef`
- 8 alliances: `qf`
- 4 alliances: `sf`
- 2 alliances: `f`

And then progress onwards as alliances get eliminated.

FRC Nexus is a tool that supports notifying teams when it's time to queue for their matches. The TBA Companion supports automating this signal and advancing through matches automatically. Additionally, for events with customized behavior, the Companion also supports importing Alliance & Playoff data from Nexus and pushing to TBA in realtime.

API Key Generation

If internet is available at your event, the companion can automatically fetch the API key for your event. If running on the FTA Toughbook, it's recommended to log in with a device account, which you can temporarily grant Lead Queuer permissions.

If internet is not available, or you would like to configure the API beforehand, you can navigate to this URL (replacing `${EVENT_KEY}` with your event's key, like `2026mawor1`): `https://frc.nexus/en/event/${EVENT_KEY}/auto-queue` and log in with an Admin / Lead Queuer account. Save the event key and API key and bring it with you to the event.

AutoQueue

To enable AutoQueue, you will need to flip the toggle on the "Nexus AutoQueue" configuration card and configure your API key. Afterwards, the companion will automatically advance through matches in Nexus. After a match score is committed in FMS, you should see the subsequent match switch to "On Field".

Alliance Selections

Warning

This should only be used when running custom playoff tournaments using Test Matches in FMS

For events with nonstandard playoff formats, you can use Nexus to power alliance selection.

- On the ”Event Setup” tab, under ”Sync Configuration”, change the Alliance Source to ”Nexus”
- After the alliance selections are entered into Nexus, go to the ”Alliances” tab and hit the ”Fetch from Nexus” button to import & push to TBA

Playoff Schedule Import

Warning

This should only be used when running custom playoff tournaments using Test Matches in FMS.

- On the ”Event Setup” tab, under ”Sync Configuration”, flip the ”Override Playoff Type” toggle and set the TBA Playoff Type to ”Custom”
- Then, set the ”Playoff Match Source” to ”Nexus + Test Match”
- Once the schedule is created in Nexus, on the ”Nexus AutoQueue” tab, hit the ”Import Playoff Matches” button to import the schedule
- Then, as you play through the schedule as Test Matches, you should see them get automatically linked to the matches from Nexus and posted to TBA

Scorekeeper Guide

These instructions will walk you through an event's flow

Qualification Matches

During Qualification matches, the data sync should happen automatically, and there is no interaction required.

Alliance Selection

After Alliance have been committed, you can manually fetch the data from the ”Alliances” tab, and then ensure they are posted.

Playoff Matches

When the playoff schedule is first activated, you may want to hit the ”sync all” button in order to ensure the initial match schedule gets posted.

Standard Format

If the playoffs are using a standard supported FMS format, then data sync will continue to happen automatically.

Non-Standard Format

If your event is using a non-standard playoff format (using "Test Match" in FMS), then you will need to manually post each match individually. After a match score is committed, a new row in the table will appear. You will need to manually assign it a TBA Match ID, and then click the "Post" button.

Refer to the match numbering guide for how to assign match codes depending on your event's playoff type.